

INTRODUCTION TO JQUERY AND AJAX

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PennApps

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WEBSITE: NO SCRIPTING



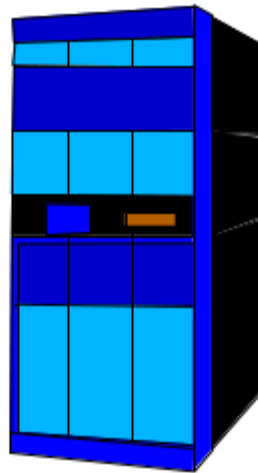
html



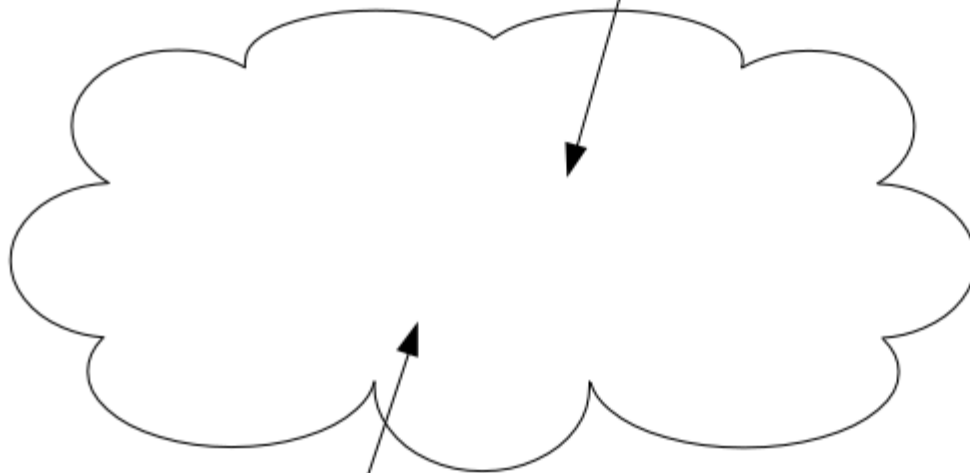
css



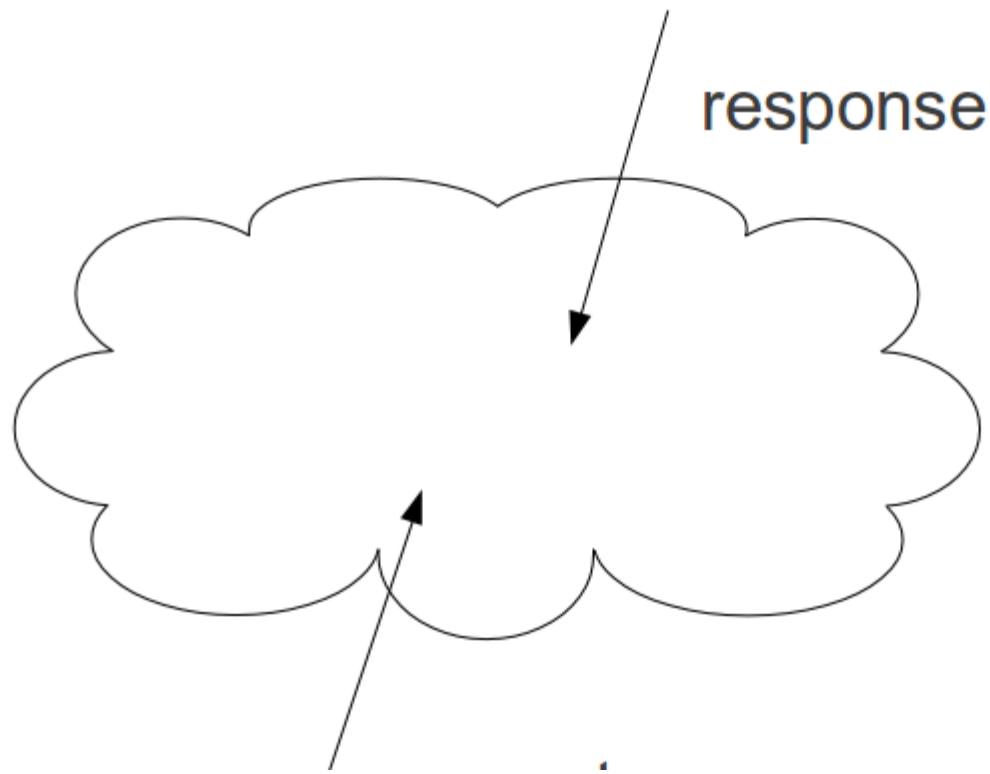
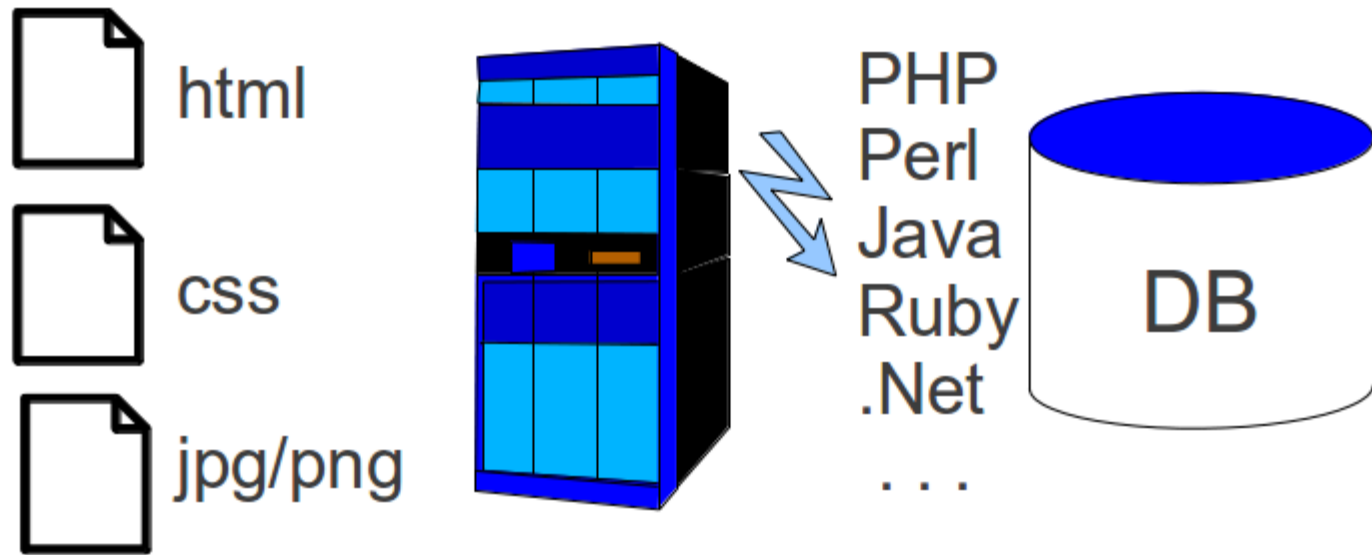
jpg/png



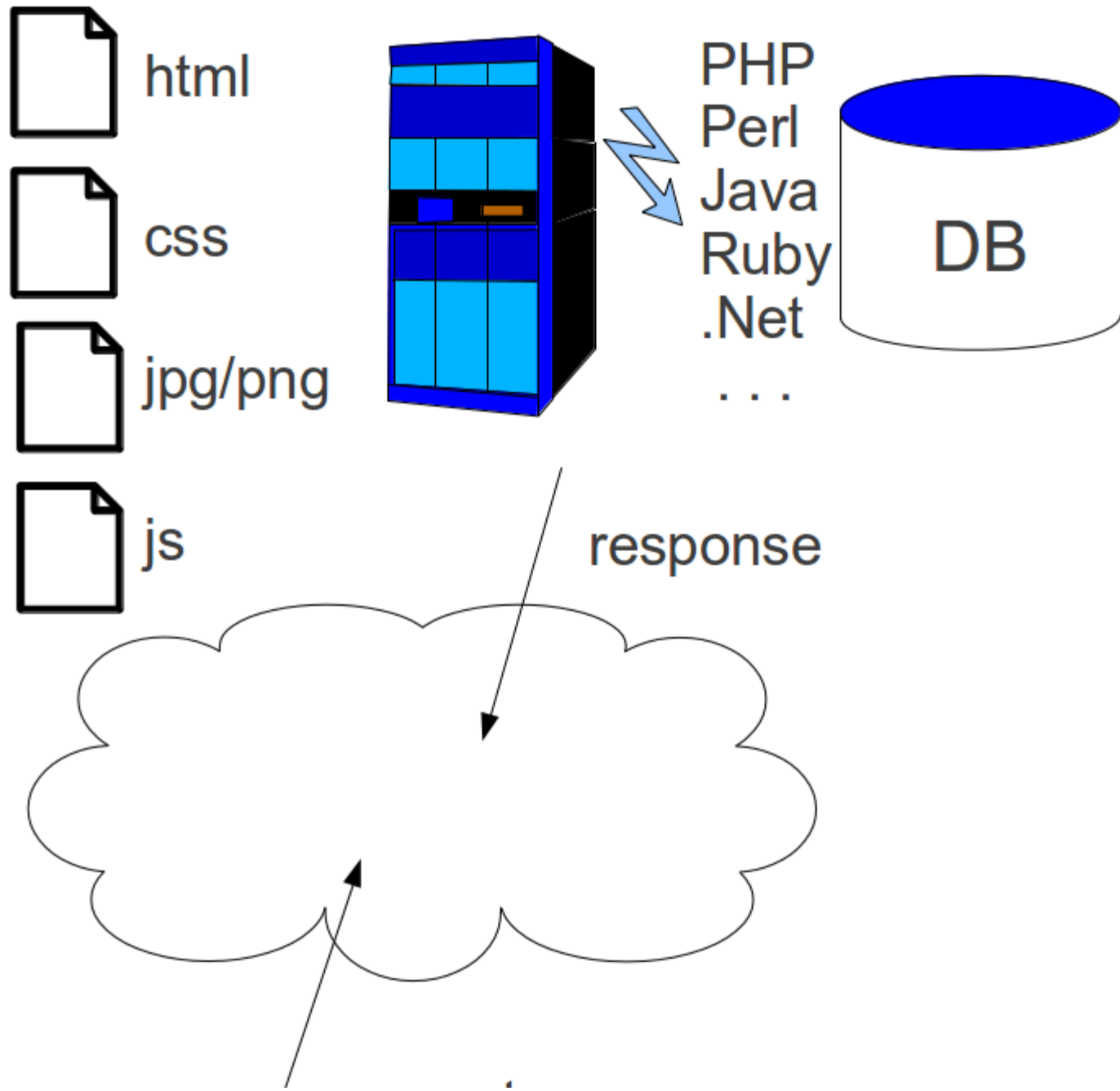
response



WEBSITE: WITH SCRIPTING



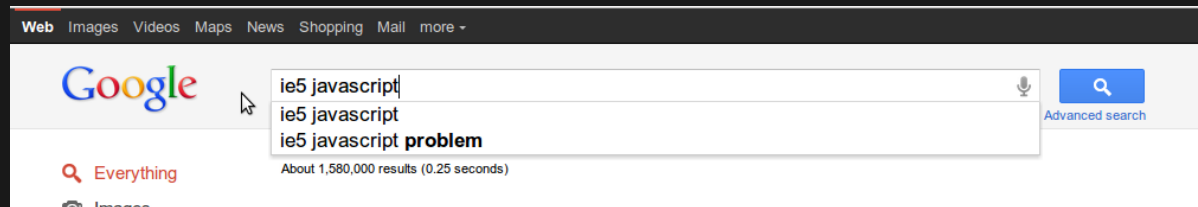
WEBSITE: WITH JAVASCRIPT



JAVASCRIPT INTRO

```
function euclidean_distance(x1, y1, x2, y2) {  
    var tmp = Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2);  
    return Math.sqrt(tmp);  
}  
  
alert(euclidean_distance(2, 2, 4, 5));
```

BROWSER INCOMPATIBILITIES

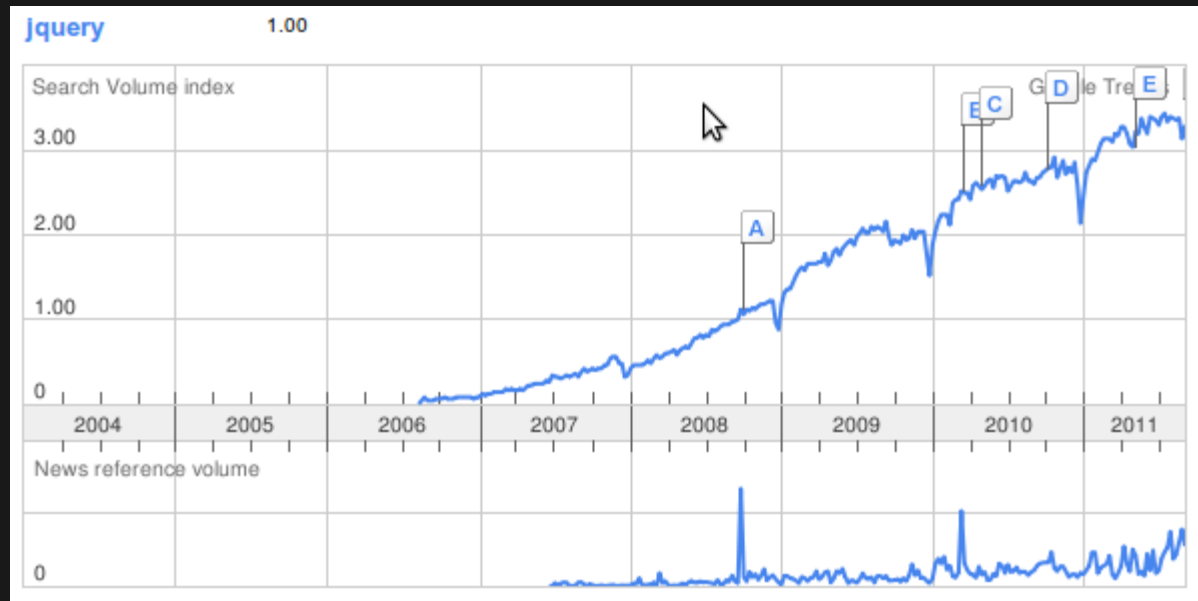


AJAX IS A PAIN

```
var xhr;
if (window.ActiveXObject) {
    xhr = new ActiveXObject("Microsoft.XMLHTTP");
} else if (window.XMLHttpRequest) {
    xhr = new XMLHttpRequest();
} else {
    throw new Error("Ajax is not supported by this browser.");
}

xhr.onreadystatechange = function() {
    if (this.readyState == 4) {
        if (this.status >= 200 &&
            this.status < 300) {
            // success
        } else {
```

JQUERY HISTORY



JQUERY PHILOSOPHY



JQUERY PHILOSOPHY

HIDE BROWSER DIFFERENCES.

DEGRADE GRACEFULLY.

USE UNOBTRUSIVE JAVASCRIPT (UJS).

```
<a href="javascript:alert('Thanks for clicking!');">click here
```

SELECT DOM ITEMS WITH CSS SYNTAX.

EVERYTHING IS A CALLBACK

CSS REVIEW

```
// Select based on the element name:
```

```
p {  
  margin-top: 0.8em;  
}
```

```
// Select based on the element's "class" attribute:
```

```
.important {  
  border: solid red 2px;  
}
```

```
// Select based on the element's "id" attribute:
```

```
#header {  
  font-family: Impact;  
  font-size: 150%;  
}
```

CALLBACKS

```
typedef void (*sighandler_t)(int);  
sighandler_t signal(int signum, sighandler_t handler);
```

CALLBACKS FOR EVENTS

```
<script type="javascript">  
  function highlight() {  
    // ...  
  }  
</script>
```

```
<a onmouseover="javascript:highlight">Click here</a>
```

FUNCTIONS ARE OBJECTS

```
setTimeout(explode, 10000);
```

or (an anonymous function):

```
setTimeout(function() {  
    alert("Time's up!");  
}, 10000);
```

FUNCTIONS ARE OBJECTS

Sub-functions:

```
function fibonacci(n) {  
  // Optimization of fibonacci(n-2) + fibonacci(n-1)  
  var results = [1, 1];  
  var _fib = function(n) {  
    if (!results[n]) {  
      results[n] = _fib(n-2) + _fib(n-1);  
    }  
    return results[n];  
  }  
  return _fib(n);  
}
```

FUNCTIONS ARE OBJECTS

Functions as constructors:

```
function Dog(theName) {  
  this.name = theName;  
  
  this.speak = function() {  
    alert(this.name + ' says "Woof!"');  
  }  
  
  this.fetch = function() {  
    // ...  
  }  
}  
  
var fido = new Dog("Fido");  
fido.speak();
```

FIRST PROBLEM: READY HANDLER

We want to run some Javascript
after the page is fully loaded.

- Necessary for UJS.
- Earliest we can access the DOM.
- This is too late:

```
window.onload = function() {  
  // ...  
}
```

FIRST PROBLEM: READY HANDLER

```
$(function() {  
  alert("Hello world!");  
});
```

FIRST PROBLEM: READY HANDLER

another example:

```
function show_login_box() {  
  if (!user_is_logged_in()) {  
    // ...  
  }  
}  
  
$(show_login_box);
```

or

```
var show_login_box = function() {  
    if (!user_is_logged_in()) {  
        // ...  
    }  
};  
  
$(show_login_box);
```

SECOND PROBLEM: MANIPULATE THE DOM

Make all our main images fade in:

```
$('#body img').fadeIn();
```

SECOND PROBLEM: MANIPULATE THE DOM

Change the html in our messages area:

```
$( '#messages' ).html( get_latest_tweets() );
```

SECOND PROBLEM: MANIPULATE THE DOM

Add a new row to a list:

```
$('#ul#todo').append("<li>Buy groceries</li>");
```

SECOND PROBLEM: MANIPULATE THE DOM

More than just CSS:

```
$('#tr:last-child').addClass('table-bottom');
```

SECOND PROBLEM: MANIPULATE THE DOM

Find all the links with an absolute URL:

```
$("a[href^='http://']").addClass('absolute-link');
```

SECOND PROBLEM: MANIPULATE THE DOM

Restrict the search based on another matched set:

```
var $game = $('#game');  
  
function kill_all_monsters() {  
    $('div.monster', $game).remove();  
}
```

SECOND PROBLEM: MANIPULATE THE DOM

Chainable:

```
function a_okay() {  
    $('#status').removeClass('red').  
                    removeClass('yellow').  
                    addClass('green');  
}
```

SECOND PROBLEM: MANIPULATE THE DOM

Lots of methods to use:

html	wrap	addClass	show
text	wrapAll	removeClass	hide
append	wrapInner	hasClass	toggle
prepend	unwrap	attr	fadeIn
appendTo	remove	css	fadeOut
prependTo	detach	offset	fadeTo

before	empty	position	slideDown
after	clone	width	slideUp
insertBefore	replaceWith	height	
insertAfter	replaceAll	data	
	val	removeData	

SECOND PROBLEM: MANIPULATE THE DOM

Or use each:

```
$(function() {  
    $('pre.source-code').each(function() {  
        parse_and_add_color(this);  
        $(this).css('border', 'dashed #aaeeff 1px');  
    });  
});
```

THIRD PROBLEM:

ADD A LISTENER

Lots of events to choose from:

click	load
dblclick	mousedown
blur	mouseenter
change	mouseleave
error	mouseout
focus	mouseover

focusin	resize
focusout	scroll
keydown	select
keypress	submit
keyup	unload

THIRD PROBLEM: ADD A LISTENER

ELECT ★ NEXT

Rick Perry



Alignment

80%

Capital punishment should be abolished.

Strongly
Disagree

Disagree

Neutral

Agree

Strongly
Agree

5 other questions
answered

Seeking
the office of:

President
of the
United
States

Election on:
September 6,
2012

THIRD PROBLEM:

ADD A LISTENER

```
<a href="/questions/56/answer?score=sd" class="answer-link">
  <div class="answer-button">Strongly<br/>Disagree</div>
</a>
<a href="/questions/56/answer?score=d" class="answer-link">
  <div class="answer-button">Disagree</div>
</a>
<a href="/questions/56/answer?score=n" class="answer-link">
  <div class="answer-button">Neutral</div>
</a>
<a href="/questions/56/answer?score=a" class="answer-link">
  <div class="answer-button">Agree</div>
</a>
<a href="/questions/56/answer?score=sa" class="answer-link">
  <div class="answer-button">Strongly<br/>Agree</div>
</a>
```

```
$(function() {  
  $('a.answer-link').click(function(event) {  
    event.preventDefault();  
    $('a.answer-link', $(this).parent()).removeClass('selected');  
    $(this).addClass('selected');  
    return false;  
  });  
});
```

FOURTH PROBLEM: CALL A REMOTE SERVER

`$.get`

`$.post`

`$.ajax`

FOURTH PROBLEM: CALL A REMOTE SERVER

```
$.ajax({  
  url: '/questions/56/answer',  
  type: 'POST',  
  data: 'score=sd',  
  dataType: 'json',  
  success: function(data, status, xhr) {  
    // update the DOM here  
  },  
  error: function(xhr, status, exception) {  
    // something went wrong  
  }  
});
```

FOURTH PROBLEM: CALL A REMOTE SERVER

or

```
$.post('/questions/56/answer', 'score=sd',  
  function(data, status, xhr) {  
    if (status == 'success') {  
      // update the DOM here  
    } else {  
      // something went wrong  
    }  
  },  
  'json');
```

FOURTH PROBLEM: CALL A REMOTE SERVER

before:

```
$('#a.answer-link').click(function(event) {  
    $.ajax({  
        url: '/questions/56/answer',  
        data: 'score=sd',  
        // ...  
    });  
});
```

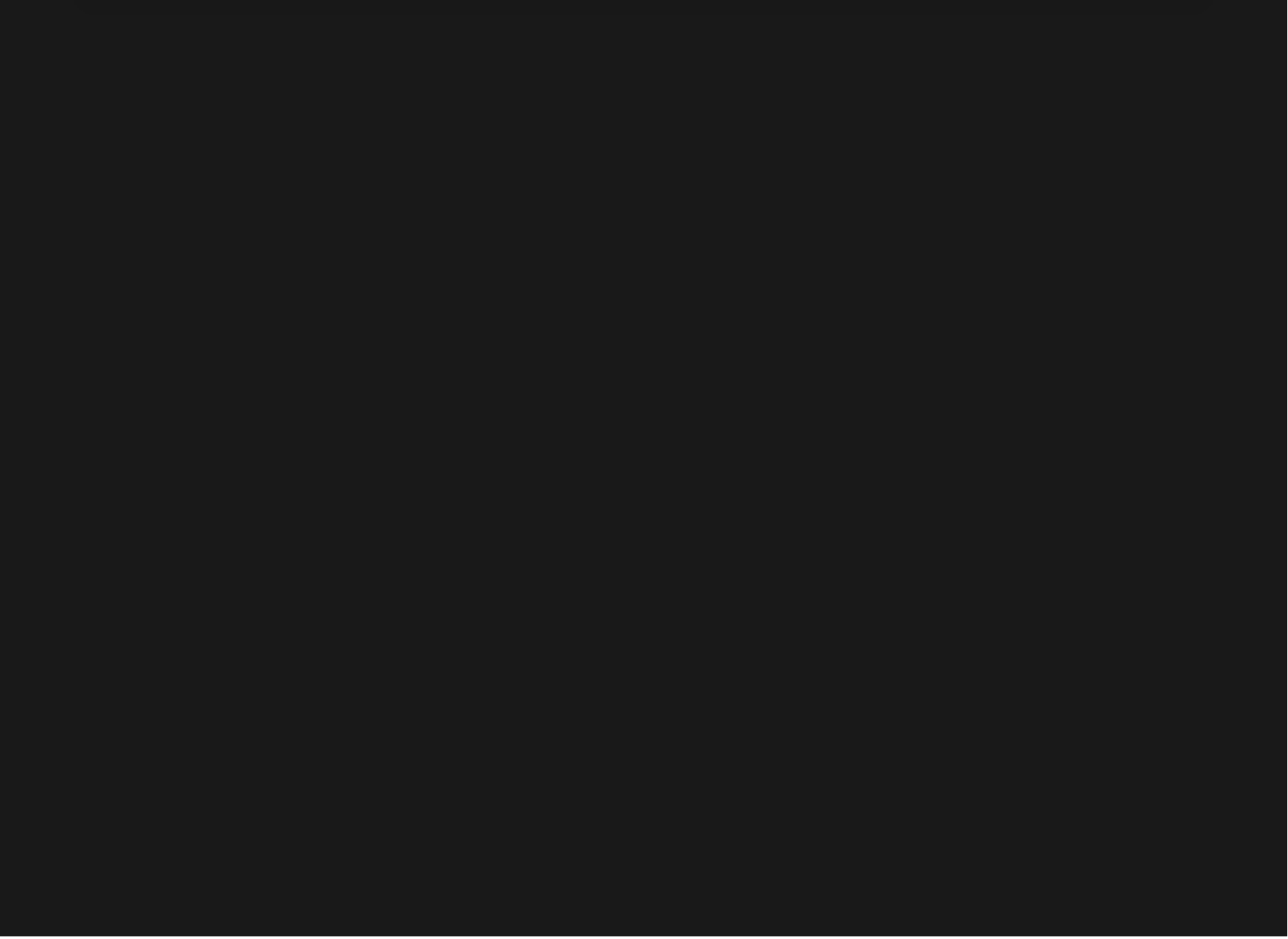
after:

```
$('#a.answer-link').click(function(event) {  
    $.ajax({  
        url: $(this).attr('href'),  
        // ...  
    });  
});
```

FOURTH PROBLEM:

CALL A REMOTE SERVER

```
$(function() {  
    $('a.answer-link').click(function(event) {  
        event.preventDefault();  
        $('a.answer-link', $(this).parent()).removeClass('selected');  
        $(this).addClass('selected');  
        $.ajax({  
            url: $(this).attr('href'),  
            type: 'POST',  
            dataType: 'json',  
            success: function(data, status, xhr) {  
                if (data.success) {  
                    $('#overall-match-score').html(data.success.score);  
                } else {  
                    $('ul#notices').html(show_errors(data.errors));  
                }  
            }  
        })  
    })  
})
```



DEMO

ELECT ★ NEXT

Rick Perry



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Disagree

Neutral

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Agree

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